LUCIDMSK CME PROGRAM

MODULE 01 — POST-TEST

Lumbar Spine MRI: Disc, Stenosis & Endplate Grading

Spine · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 **On sagittal T2, a lumbar disc shows inhomogeneous gray-black signal, lost NP/AF distinction, and moderately decreased height. Pfirrmann grade:**

- A Grade II
- B Grade III
- C Grade IV
- D Grade V

2 **A disc herniation where the dome is WIDER than the neck, remaining in continuity with the parent disc, is classified as:**

- A Broad-based bulge
- B Protrusion
- C Extrusion
- D Sequestration

3 **Axial T2 shows no CSF signal but nerve roots remain individually distinguishable. Schizas grade:**

- A Grade A
- B Grade B
- C Grade C
- D Grade D

4 **Which Modic type(s) are FDA-cleared targets for basivertebral nerve (BVN) ablation?**

- A Type 1 only
- B Type 2 only
- C Types 1 and 2
- D Types 1, 2, and 3

5 **Sagittal T1 shows completely obliterated perineural fat with nerve root deformation at L4-5 right foramen. Lee grade:**

A	Grade 0
B	Grade 1
C	Grade 2
D	Grade 3

6 **L4 anterolisthesis on L5 measures 60% anterior slip. Meyerding grade:**

A	Grade II
B	Grade III
C	Grade IV
D	Grade V (spondyloptosis)

7 **A subarticular (paracentral) disc herniation at L4-5 most commonly compresses which nerve root?**

A	L3
B	L4
C	L5
D	S1

8 **Modic Type 1 is best characterized by which MRI signal pattern?**

A	T1 hypointense, T2 hypointense
B	T1 hyperintense, T2 suppressed on STIR
C	T1 hypointense, T2/STIR hyperintense
D	T1 hyperintense, T2 hyperintense

9 **A disc fragment has lost all continuity with the parent disc and migrated caudally. Best term:**

A	Protrusion with migration
B	Extrusion
C	Sequestration
D	Foraminal herniation

10 Key distinguishing feature of discitis vs. Modic Type 1 on MRI:

A	Presence of T2 endplate signal alone
B	Disc height loss + T2 disc signal + paraspinal/epidural soft tissue
C	Subchondral sclerosis on T1
D	Enhancement of endplates post-contrast

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	C	C	C	B	C	C	C	B

Q1 **On sagittal T2, a lumbar disc shows inhomogeneous gray-black signal, lost NP/AF distinction, and moderately decreased height. Pfirrmann grade:** **Answer: C**

A Grade II

B Grade III

✓ **Grade IV**

D Grade V

EXPLANATION Grade IV = inhomogeneous gray-black signal, lost NP/AF distinction, moderately decreased height. Grade V = completely collapsed disc space. Grade III has unclear (not lost) NP/AF.

Q2 **A disc herniation where the dome is WIDER than the neck, remaining in continuity with the parent disc, is classified as:** **Answer: C**

A Broad-based bulge

B Protrusion

✓ **Extrusion**

D Sequestration

EXPLANATION Extrusion = dome wider than base (neck), maintained continuity. Protrusion = base wider than dome. Sequestration = lost continuity. Bulge = >180° symmetric extension, not a true herniation.

Q3 **Axial T2 shows no CSF signal but nerve roots remain individually distinguishable. Schizas grade:** **Answer: C**

A Grade A

B Grade B

✓ **Grade C**

D Grade D

EXPLANATION Schizas C = no CSF but roots still identifiable as separate structures. Grade D = no CSF AND roots indistinguishable (solid block of tissue). Grade B = inhomogeneous but CSF still present.

Q4 **Which Modic type(s) are FDA-cleared targets for basivertebral nerve (BVN) ablation?** **Answer: C**

A Type 1 only

B Type 2 only

✓ **Types 1 and 2**

D	Types 1, 2, and 3

EXPLANATION Types 1 and 2 are the FDA-cleared indications for Intracept BVN ablation. Type 3 (sclerotic) is not a target. Type 1 has the strongest published pain response evidence.

Q5 **Sagittal T1 shows completely obliterated perineural fat with nerve root deformation at L4-5 right foramen. Lee grade:** **Answer: C**

A	Grade 0
B	Grade 1
✓	**Grade 2**
D	Grade 3

EXPLANATION Lee Grade 2 = completely obliterated fat + nerve root deformation. Grade 1 = partially obliterated without deformation. Grade 3 = nerve root entirely invisible within the foramen.

Q6 **L4 anterolisthesis on L5 measures 60% anterior slip. Meyerding grade:** **Answer: B**

A	Grade II
✓	**Grade III**
C	Grade IV
D	Grade V (spondyloptosis)

EXPLANATION Grade III = 51–75% slip. Grade II = 26–50%. Grade IV = 76–100%. Grade V = >100% (spondyloptosis, complete displacement). Grades III+ usually require instrumented fusion.

Q7 **A subarticular (paracentral) disc herniation at L4-5 most commonly compresses which nerve root?** **Answer: C**

A	L3
B	L4
✓	**L5**
D	S1

EXPLANATION Subarticular herniations compress the TRAVERSING root, one level below the disc level. L4-5 subarticular → L5 root. A FORAMINAL herniation at L4-5 compresses the EXITING L4 root.

Q8 **Modic Type 1 is best characterized by which MRI signal pattern?** **Answer: C**

A	T1 hypointense, T2 hypointense
B	T1 hyperintense, T2 suppressed on STIR
✓	**T1 hypointense, T2/STIR hyperintense**
D	T1 hyperintense, T2 hyperintense

EXPLANATION Type 1 = T1↓ hypointense, T2/STIR↑ hyperintense — active inflammatory vascular fibrous tissue. Type 2 = T1↑, STIR suppressed (fatty marrow). Type 3 = both hypointense (sclerosis).

Q9 **A disc fragment has lost all continuity with the parent disc and migrated caudally. Best term:** **Answer: C**

A Protrusion with migration

B Extrusion

✓ **Sequestration**

D Foraminal herniation

EXPLANATION Sequestration = free fragment with NO continuity to parent disc. Extrusion retains continuity. Migration describes caudal travel. Always report direction and estimated distance.

Q10 **Key distinguishing feature of discitis vs. Modic Type 1 on MRI:** **Answer: B**

A Presence of T2 endplate signal alone

✓ **Disc height loss + T2 disc signal + paraspinal/epidural soft tissue**

C Subchondral sclerosis on T1

D Enhancement of endplates post-contrast

EXPLANATION Discitis shows disc HEIGHT LOSS, T2 hyperintensity of the DISC ITSELF, and paraspinal/epidural extension. Modic 1 preserves disc height and lacks soft tissue spread. Both can show endplate enhancement.